AI Intelligent Constituency Mission — Public Data & API Map

Every dataset and API below is reachable from an authorised Kerala or Government of India portal. Sources are linked to the twin-model layers they populate and the AICM delivery modules they feed. The Mermaid diagram at the end shows the full hierarchy from government data source → twin layer → module → feature.

> **Access reality:** "Open download" does not always mean "open API." KSDI provides OGC WFS/WMS but bulk vector downloads may require registration[reference:0][reference:1]. Sanchaya and Sanketham expose web portals and payment endpoints but not documented public REST APIs for bulk extraction[reference:2][reference:3]. IMD provides an API management platform but some endpoints require IP whitelisting[reference:4][reference:5]. The portal URLs below are the authoritative starting points; bulk extraction should be routed through IKM or KSDI where required.

01 Data Source & API Register

#	DATA POINT / DATASET	GOVERNMENT SOURCE	ACCESS TYPE	TWIN LAYER	MODULE	API / ENDPOINT NOTE
1	Panchayat & ward boundaries, hydrography, forest, roads	KSDI	OGC WFS/WMS	L1 Spatial Base	M1 M4 M5	Open WFS endpoint; example vector service at `ksdi.kerala.gov.in/erdas-apollo/vector/…`[reference:6]
2	Cadastral parcels + land classification (Nilam / Purayidom)	Ente Bhoomi · ReLIS	Portal / view	L2 Cadastral	M3 M7	Digitized cadastral maps and land data bank registers; bulk download requires departmental clearance
3	Building footprints + property tax attributes	Sanchaya · IKM	Portal	L3 Built Env.	M2 M3 M6	Property tax assessment and payment portal; API-based access available only through IKM coordination[reference:7]
4	Road network — NH / SH / MDR + local roads	PWD GeoJSON · KSDI	GeoJSON download	L3 Built Env.	M2 M4	Community-maintained PWD extracts; authoritative layer via KSDI WFS
5	Utility network — transformers, HT lines	KSEB via KSDI	WFS via KSDI	L3 Built Env.	M2	KSEB spatial data sharing sanctioned through KSDI geoportal[reference:8]
6	Ward demographics, voters, SHG membership	OpenDataKerala · Dataful · DES	CSV / GeoJSON	L4 Socio-Demo.	M5 M6	OpenDataKerala LSG dashboard source data; Dataful SHG and election datasets
7	Crop area & production (EARAS)	DES	Dashboard API	L8 Natural Res.	M7	DES interactive dashboard integrated through API with custom data tables[reference:9]; EARAS analytics at data portal
8	Groundwater quality & aquifer level	NWDP · CGWB	CSV / API	L8 Natural Res.	M7	NWDP exposes groundwater level datasets in CSV and API format; telemetry hourly and manual quarterly[reference:11]
9	Health facility HR & infrastructure	DES Medical Institutions · e-Health Dashboard	Portal / dashboard	L5 Service Infra.	M5	e-Health dashboard publicly available; NHP hospital directory CSV at `info.nhp.gov.in/api/hospital-directory.csv`[reference:12]
10	School enrolment & staffing	UDISE+	REST API	L5 Service Infra.	M5	UDISE+ Developer User Portal provides school APIs by UDISE code, district/block lists, and modified-date queries[reference:14]

#	DATA POINT / DATASET	GOVERNMENT SOURCE	ACCESS TYPE	TWIN LAYER	MODULE	API / ENDPOINT NOTE
11	GPDP projects & expenditure	eGramSwaraj · Samvedhitha	Portal / PDF	L6 Plan & Finance	M1 M8	GPDP plans and reports downloadable as PDF; no documented public REST API for bulk project-line extraction[reference:15]
12	PDI composite score (9 themes)	PDI Portal	Portal / database	L6 Plan & Finance	M1 M5	MoPR NIC online database management application for PDI; data porting through API, Excel or CSP format[reference:16]
13	SDG indicators (localised)	Kerala SDG Dashboard	Dashboard	L6 Plan & Finance	M1 M5	State/district/local functionaries track SDG progress; real-time dashboards for monitoring[reference:17] [reference:18]
14	Hazard maps + Orange Book protocols	KSDMA · KILA DMP	PDF / raster	L7 Risk & Resilience	M4	Flood hazard probability rasters available; inundation depth reclassifiable; landslide and drought zones in KSDMA and DDMPs[reference:19]
15	Grievance reports + citizen feedback (geo-tagged)	CMO Grievance Portal · Panchayat system	Secure API (internal)	L3 / L6 Overlay	M1 M6	CMO grievance system uses secure APIs for information exchange; public bulk API not exposed[reference:20]

02 AICM-Specific Additional Sources

These sources are required for AICM components that go beyond the Panchayat Twin — industrial intelligence, innovation index, disaster prediction, and citizen assistant.

#	DATA POINT / DATASET	GOVERNMENT SOURCE	ACCESS TYPE	AICM COMPONENT	API / ENDPOINT NOTE
16	Real-time weather observations, forecasts, warnings, nowcast	IMD API Management Platform	REST API	Disaster Preparedness	Unified gateway for city forecasts, nowcast, rainfall, marine bulletins, warning services; some endpoints require IP whitelisting[reference:21][reference:22]
17	Startup registry, sector, stage, location	Kerala Startup Mission — Kernal API	REST API	Industrial Intelligence · CII	Startup search endpoint at `startups.startupmission.in/api/v1/startup/search`[reference:23]
18	MSME unit registration, district-wise industrial data	State EoDB Dashboard · Directorate of Industries	Dashboard	Industrial Intelligence	Directorate of Industries and Commerce dashboard; DIC login-based field data capture[reference:24][reference:25]
19	Patent filings and IP data	IP India · KSCSTE PIC	No public API	Constituency Innovation Index	No official public REST/SOAP API from IP India; public search only; WIPO API Catalog lists some IP office APIs[reference:26][reference:27]
20	OpenStreetMap features — schools, hospitals, amenities, roads	OpenStreetMap Overpass API	Overpass API	Citizen Assistant · Facility Finder	Query by geocode area or bounding box; export GeoJSON; community-maintained Kerala extracts on GitHub[reference:28]

#	DATA POINT / DATASET	GOVERNMENT SOURCE	ACCESS TYPE	AICM COMPONENT	API / ENDPOINT NOTE
21	Bhuvan Panchayat thematic layers — LULC, amenities, infrastructure	Bhuvan Panchayat	WMS / shapefile	Digital Twin · Facility Gap	OGC WMS/WMTS services; direct shapefile download available; 1:10k thematic products[reference:30]
22	Census population, households, literacy, SC/ST	Census of India · censusindia REST API	REST API	Social Well-being · Digital Twin	Community REST API wraps Census 1901–2011 with boundaries and projections; ArcGIS FeatureServer also available[reference:31][reference:32]
23	Kerala OGD — department-published tables, APIs, civic datasets	Kerala OGD (data.gov.in instance)	CKAN API	Cross-cutting	State instance of national OGD platform; ~10,000 resources from 30+ agencies; CKAN-style API discovery[reference:33]
24	Building permit applications and approvals	Sanketham	Portal	Development Copilot · Permit Pre-check	Building permit portal integrated with Soochika and Saankhya; no documented public bulk API[reference:35][reference:36]

03 Hierarchy: Government Source → Twin Layer → Module → Feature

The diagram reads top-to-bottom. Each government source feeds one or more twin layers; each layer feeds delivery modules; each module delivers features to the President or Citizen.



Coverage statement. This register consolidates **24 government data sources**, **8 twin-model layers**, **8 delivery modules**, and **43 features** (21 President, 16 Citizen, 6 Shared) plus AICM-specific extensions. Sources span KSDI, Ente Bhoomi, ReLIS, Sanchaya, IKM, PWD, KSEB, DES, UDISE+, eGramSwaraj, Samvedhitha, PDI, Kerala SDG Dashboard, KSDMA, KILA DMP, IMD, KSM Kernal API, EoDB Dashboard, IP India, OSM Overpass, Bhuvan Panchayat, Censusindia, Kerala OGD, CMO grievance, and Sanketham.

API reality check. Only a subset of sources expose documented, machine-readable public APIs: KSDI (WFS/WMS), UDISE+ (REST), IMD (REST with possible IP whitelisting), KSM Kernal (REST), OSM Overpass (API), Censusindia (REST), NWDP (CSV/API), and Kerala OGD (CKAN). Sanchaya, Sanketham, eGramSwaraj, PDI portal, and CMO grievance are portal-first; bulk extraction requires IKM/KSDI coordination or departmental clearance. IP India has no public API. Plan for a hybrid extraction strategy — API where available, structured portal scraping or PDF parsing where not, and always with a freshness stamp.